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How Exactly Do Network Relationships Pay Off? The Effects of Network Size and Relationship Quality on Access to Start-Up Resources

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Entrepreneurs are expected to profit from network relationships. Research addressing the link between entrepreneurs' network characteristics and their performance, however, has so far produced inconclusive results. In an attempt to explain these inconsistencies, we investigate the resource returns connected to network size and relationship quality. Based on a sample of 379 nascent entrepreneurs, we find that increasing network size and relationship quality results in diminishing marginal returns in terms of access to financial capital, knowledge and information, and additional business contacts. Additionally, we observe that the returns vary strongly by resource type.

Introduction

Since the 1980s, researchers have argued that entrepreneurs' networks—defined as the set of relationships or contacts held by entrepreneurs—are important for their success in all stages of the entrepreneurial process (e.g., Aldrich, Rosen, & Woodward, 1987; Birley, 1985; Hoang & Antoncic, 2003; Stuart & Sorenson, 2007). The rationale underlying this claim is a rather simple one: Network relationships are expected to provide access to needed resources at attractive terms (Adler & Kwon, 2002; Uzzi, 1999). Consequently, it is assumed that entrepreneurs, who usually have to rely on external resources

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to successfully establish and develop their new ventures, profit from developing and maintaining network relationships (e.g., Batjargal, 2003; Elfring & Hulsink, 2007; Kim & Aldrich, 2005; Liao & Welsch, 2005).

Empirical research addressing the connection between network characteristics and entrepreneurs' success, however, has so far produced inconclusive results. On the one hand, there are several studies in the field showing a positive effect of larger or more diverse networks (e.g., Raz & Gloor, 2007; Stam & Elfring, 2008) and of stronger network relationships (e.g., Lee & Tsang, 2001). Other studies addressing the same variables indicate no significant link, or sometimes even a negative effect (e.g., Aldrich & Reese, 1993; Batjargal, 2005, 2010; Johannisson, 1996; Watson, 2007).

To help explain these conflicting findings, this paper provides additional insights into the bases on the widely proposed performance effect of network ties, which have so far received very limited attention from empirical researchers in the field (Street & Cameron, 2007; Stuart & Sorensen, 2007). Specifically, we develop and test the hypotheses that extending a network and increasing relationship quality lead to positive but diminishing returns in terms of access to financial capital, knowledge and information, and additional business contacts. Based on a sample of 379 nascent entrepreneurs from Germany, our results provide strong support for our hypotheses. Additionally, they reveal that the size of the network and the level of relationship quality needed to access different types of resources vary considerably.

Based on these insights, we think that our study contributes to entrepreneurship research by pointing to two complementary explanations for the previous inconclusive findings on the relationship between entrepreneurs' network characteristics and their success. First, and especially in view of the fact that both activities likely come with considerable opportunity costs in time, our finding that extending a network and increasing the quality of network relationships both result in diminishing resource returns indicates that one should not expect the ultimate performance effects of increasing network size and relationship quality to be linear. Second, the observation that the network size as well as the level of relationship quality needed to gain access to different types of resources vary considerably, as well as the tentative explanation we provide for this effect, points to the fact that the type of resource support sought may significantly affect the value of larger networks and stronger network relationships. Finally, considering the fact that the variables addressed in our study are, at least to some extent, under the control of entrepreneurs, we believe that our findings have considerable practical implications.

Theory and Hypotheses

For around 30 years, researchers have argued that network relationships are important for entrepreneurs' success in all stages of the founding process (e.g., Aldrich et al., 1987; Birley, 1985; Hoang & Antoncic, 2003; Stuart & Sorenson, 2007).

As depicted in Table 1, however, prior research addressing the connection between entrepreneurs' network characteristics and their success has produced ambiguous results. Broadly, research focusing on the relationship between entrepreneurs' network characteristics and their success may be divided into two camps—one focusing on the effects of relational, the other focusing on the effects of structural, network characteristics (see, e.g., Granovetter, 1992; Hoang & Antoncic, 2003).

Table 1

Findings on Network Characteristics and Entrepreneurs' Success

Camp	Variable	Authors	Findings
Relational	Tie strength–strong vs. weak ties	Batjargal (2003)	Entrepreneurs profit from acquaintance-based relationships, while friendship-based strong ties may negatively affect their performance.
		Davidsson and Honig (2003)	Strong and weak tie support is a valid predictor for advancing through the start-up process.
		Brüderl and Preisendorfer (1998)	Network support increases the chances of new venture survival and growth. Support from strong ties seems more important than support from weak ties.
	Tie strength–relationship quality	Hansen (1995)	Entrepreneurs who interact more frequently with network partners are more successful in terms of venture growth.
		Lee and Tsang (2001)	There is a significant positive relationship between entrepreneurs' frequency of interaction with network partners and the growth of small firms.
		Aldrich et al. (1987)	(Nascent) entrepreneurs spending more time on network contacts are more likely to found a business and make profit.
		Aldrich and Reese (1993)	There is no significant relationship between the time entrepreneurs spend on developing and maintaining business contacts and venture survival.
		Johannisson (1996)	There is no significant relationship between tie strength and the performance of (nascent) entrepreneurs.
		Watson (2007)	The frequency with which venture owners seek advice from network sources is, to some extent, positively related to performance, but afterward even hinders venture survival and growth.
Structural	Network position	Stam and Elfring (2008)	Bridging ties has a statistically significant positive relationship with sales growth and performance relative to competitors.
		Batjargal (2003)	Network heterophily, in terms of the diversity of contacts, has no significant effect on entrepreneurs' profit margin.
		Batjargal (2005)	The diversity of entrepreneurs' network contacts has no impact on their new ventures' revenue growth.
		Batjargal (2007)	The performance of entrepreneurs with more start-up experience is negatively affected by structural holes in their network.
		Batjargal (2010)	Structural holes have a negative effect on a new venture's profit growth.
		Hansen (1995)	The interconnectedness of network members is significantly positively related to venture growth.
	Network size	Raz and Gloor (2007)	The size of informal interfirm networks has a positive impact on new venture survival.
		Baum, Calabrese, and Silverman (2000)	New ventures profit from having a larger alliance network at founding.
		Hansen (1995)	The size of an entrepreneur's network has a positive impact on new venture growth.
		Aldrich and Reese (1993)	There is no significant relationship between entrepreneurs' network size and business survival.
		Johannisson (1996)	There is no distinct pattern between network size and the performance of (nascent) entrepreneurs.
		Batjargal (2003)	The size of entrepreneurs' networks and their profit margins are not significantly related.
		Batjargal (2005)	The size of an entrepreneur's personal network is not related to growth of his/her venture's revenue.

Relationship Quality and Entrepreneurs' Success

The more relational-oriented stream of network research in entrepreneurship is mainly based on Granovetter's (1973) classic work on the relevance of weak ties. Granovetter claimed that weak ties, such as connections to business network members, as opposed to the strong ties typical of family relations or friendships, may be especially beneficial because they provide access to diverse resources not available within an individual's immediate social circles. Following this line of thought, many researchers have empirically analyzed how entrepreneurs' performance is influenced by support from strong ties on the one hand and weak ties on the other. As depicted in Table 1, the results generated by this research are ambiguous.

From a theoretical point of view, these inconclusive findings may be explained when taking into account that the underlying reasoning for the "strength of weak ties" (Granovetter, 1973, p. 1360) is based on the assumptions that (1) relationships to family members and friends are, in general, stronger than are those to business contacts, and that (2) the strength of relationships between a focal actor and his or her network partners is positively correlated to network partners' interrelatedness, which is in turn expected to have a negative impact on the variety of resources available via network partners. Empirically, however, both of these claims are not self-evident. Steier and Greenwood (2000), for example, have shown that business relationships may turn into very strong bonds, and more recent research results show no significant connection between the strength of network ties and network partners' interconnectedness (Bhagavatula, Elfring, van Tilburg, & van de Bunt, 2010).

However, even relational-oriented network research that defines relationship quality independently from the identity of networks partners, that measures the variable based on the frequency of interaction or the time spent on a relationship, and that explains potential positive performance effects of higher relationship quality by an increase in network partners' willingness to provide help and support (Adler & Kwon, 2002; Krackhardt, 1992) has so far also produced inconclusive findings.

Network Structure and Entrepreneurs' Success

As depicted in Table 1, studies addressing the effects of structural network characteristics on entrepreneurs' performance also fail to provide a clear picture. Structure-related research on entrepreneurs' networks may again be divided into two subgroups—research addressing the effects of network size and research analyzing the impact of network position (Hoang & Antoncic, 2003).

The discussion of the relevance of network position on performance is dominated by two opposing views. On the one hand, there is a stream of research based on Coleman's (1988, 1990) claim that dense networks with highly interconnected actors should be beneficial for the development of trust among network members, which in turn facilitates resource exchange. In contrast, the research perspective best represented by the work of Burt (1982, 1997, 2005) highlights the advantages of sparse networks with disconnected network partners. From his point of view, a focal actor holding a bridging position in a network—meaning that he or she connects otherwise unconnected network alters—is able to access more diverse resources and reap the benefits from brokering resource flows within the network. Even though this latter perspective in particular has been widely recognized in entrepreneurship research to date, there are studies providing confirming evidence for both positions, and other research results that show no significant performance effects at all.

Even more often than addressing entrepreneurs' network position, researchers in the field have focused on the question of whether entrepreneurs profit from larger networks (Hoang & Antoncic, 2003; Jack, 2010). Again, as reflected in Table 1, the results reported for the relationship between the size of entrepreneurs' networks and their performance in terms of successfully founding and developing a new venture are not entirely conclusive.

Network Relationships and Entrepreneurs' Resource Access

One potential explanation for the fact that research addressing the relationship between entrepreneurs' networks and their performance or success has so far not produced a clear picture may be that the relationship between network characteristics and resource access, which is expected to provide the basis for a potential positive performance impact of entrepreneurs' network relationships (see, e.g., Bhagavatula et al., 2010; Hoang & Antoncic, 2003; Jack, 2010; Stam & Elfring, 2008), is under-investigated so far.

In fact, the link between networks and resource access has only been analyzed by very few studies (Street & Cameron, 2007; Stuart & Sorensen, 2007). There are some qualitative studies indicating that the quality of network relationships may have an impact on resource access (Jack, 2005; Steier & Greenwood, 1995, 2000), and two quantitative studies underscoring this idea by revealing a positive relationship between the quality of relationships between entrepreneurs and external investors and entrepreneurs' access to financial capital (Batjargal & Liu, 2004; Kwon & Arenius, 2010). Moreover, there are some studies showing that the reputation and legitimacy of entrepreneurs' network members may positively influence their access to financial capital (Gulati & Higgins, 2003; Honig, Lerner, & Raban, 2006; Stuart, Hoang, & Hybels, 1999), and a single study providing evidence for the notion that network sparseness may negatively influence access to financial capital (Bhagavatula et al., 2010).

Even though we definitely recognize the contributions made by this prior research, we find that there is a dearth of research systematically addressing how the characteristics of entrepreneurs' networks affect their access to needed resources. Assuming that shedding more light on this connection may help explain the inconclusive findings on entrepreneurs' network characteristics and success, we do so in this paper, focusing on nascent entrepreneurs, who are expected to particularly depend on their networks to gather the resources necessary for founding a new venture (see, e.g., Aldrich, 1999; Hanlon & Saunders, 2007; Yoo, 2000).

Characteristics of Nascent Entrepreneurs' Networks and Access to Start-Up Resources

Nascent entrepreneurs are individuals actively engaged in founding a new venture (Aldrich, 1999; Reynolds, Carter, Gartner, & Greene, 2004). To master the transition of firm birth, they have to complete many different tasks and need plentiful and diverse resources (Aldrich; Hanlon & Saunders, 2007; Yoo, 2000). Consequently, it is expected that mobilizing external resources is a key to their success, and network ties are considered to be a major source for them to do so (Bowey & Easton, 2007; Casson & Giusta, 2007; Greve & Salaff, 2003).

The variety of resources that network relationships are expected to provide to nascent entrepreneurs is considerable. According to Aldrich et al. (1987) and Aldrich (1999), for example, nascent entrepreneurs should be able to obtain through their networks resources such as money, product ideas, and information. Quite similarly, Yoo (2000) posits that nascent entrepreneurs may mobilize knowledge, information, and financial capital through their network contacts. Combining these results with other insights in the field, the resources that nascent entrepreneurs are expected to obtain via network relationships can be put into three categories: (1) financial capital (Casson & Giusta, 2007; Hanlon & Saunders, 2007); (2) information and knowledge (Chell & Baines, 2000; Liao & Welsch, 2005; Yoo); and (3) access to additional business-relevant contacts, such as potential customers and consultants (Aldrich; Aldrich et al.).

Based on this notion, we will subsequently develop and later on test detailed hypotheses on how network size and relationship quality—two network variables that prominently represent the two different camps in network research described above—are related to nascent entrepreneurs' resource access.

Extending Network Size and Resource Access. As proposed by many theorists in the field, larger networks are expected to pay off because network contacts provide access to resources, and the size of a network indicates how many different resource holders nascent entrepreneurs can potentially rely on when trying to establish their ventures (Aldrich et al., 1987; Liao & Welsch, 2005). Besides mere quantity, however, the variety of resources available through a network is also expected to increase with size (Grandi & Grimaldi, 2003; Greve & Salaff, 2003). The reasoning behind this proposition is that because people differ with respect to their stocks of knowledge and other resources, including more people in a network should also increase the variety of resources available to nascent entrepreneurs (Greve & Salaff).

In line with findings from Chell and Baines (2000), Grandi and Grimaldi (2003), and Batjargal (2003), however, this effect is not limited to accessing information and knowledge. When starting a new venture, nascent entrepreneurs have a network of social contacts comprising family members, friends, and work-related ties (Brüderl & Preisendorfer, 1998; Elfring & Hulsink, 2003). Being willing to support a nascent entrepreneur without expecting much in return, such pre-existing ties are—especially in the beginning of the entrepreneurial process—a valuable asset. In the course of the process of new venture development, however, nascent entrepreneurs will almost inevitably face resource needs that cannot be satisfied by their pre-existing network contacts (Hite & Hesterly, 2001; Larson & Starr, 1993). In such a situation, by expanding their network and including additional contacts such as industry experts, lawyers, consultants, and business angels, nascent entrepreneurs may increase their chances of getting access to the different resources needed, such as information and knowledge, contacts to potential clients or suppliers, and even financial capital.

However, we do not expect the probability of receiving the resources needed to increase at a constant rate when a network is extended. Whereas adding more partners to a small network will significantly enhance the probability that new resources will become available, this effect will diminish as a network grows. When a network is already of considerable size, adding more contacts will much more likely increase network "redundancy" since additional ties will more probably duplicate resource access. Still, however, extending their networks may be beneficial for nascent entrepreneurs because it reduces their dependency on specific network contacts (Reagans & Zuckerman, 2008). Consider a situation in which a nascent entrepreneur has only a single contact providing access to financial capital. In this situation, the nascent entrepreneur depends on that one contact for capital. When increasing network redundancy, however, a nascent entrepreneur increases the probability that he or she has access to the resource in question at affordable terms. Again, however, we expect the relevance of this effect to decrease with network size. When network redundancy has already reached a sufficient level, adding additional ties will no longer have a significant positive effect in terms of reducing dependency.

Combining these arguments, we thus expect that when nascent entrepreneurs expand their network, it positively influences their chances of accessing needed resources, but the marginal returns diminish as additional network ties are added. Consequently, we hypothesize that the relationship between the size of nascent entrepreneurs' networks and their access to start-up relevant resources, such as financial support, knowledge/information, and additional business contacts, is positive but concave:

> **Hypothesis 1:** There will be a positive but concave relationship between the size of a nascent entrepreneur's network and access to start-up relevant resources via network contacts.

Increasing Relationship Quality and Resource Access. In a manner similar to the effect described above, we expect nascent entrepreneurs to also profit from spending more time on a given number of network contacts. Specifically, and in line with the findings of Duchesneau and Gartner (1990), who observe that successful entrepreneurs spend more time on communicating with network partners, and Elfring and Hulsink (2003), who argue that spending time to develop strong bonds with network contacts is crucial for entrepreneurial success, we hypothesize that nascent entrepreneurs who spend more time on a given number of network contacts will enhance their chances of getting access to relevant resources, such as knowledge and information, financial capital, and additional business contacts.

The general existence of a relationship between two individuals does not mean that they will grant each other access to resources. Instead, network relationships have to have a certain quality to make this happen (Adler & Kwon, 2002; Krackhardt, 1992; McFadyen & Cannella, 2004; Portes, 1998), and a significant amount of time has to be invested to develop a relationship of such quality (Aldrich & Reese, 1993; Chunyan, 2005; Elfring & Hulsink, 2003). In spending time together, network partners develop stronger bonds, increasing their mutual trust and commitment, and thus their motivation to help one another by granting access to their resources at favorable terms (Krackhardt; McFadyen & Cannella; Steier & Greenwood, 2000). Additionally, there is another benefit when network partners spend time together: When doing so, individuals develop norms guiding their interactions and a shared understanding that facilitates resource exchange in general, and the transfer of knowledge and information in particular (McFadyen & Cannella; Uzzi, 1997). In view of these arguments, we expect nascent entrepreneurs who spend more time on developing network relationships with a higher quality to increase their chances of accessing resources needed for setting up a new venture through their network.

Again, however, we do not expect the relationship between relationship quality and the availability of resources to be linear. Although increasing the amount of time dedicated to a network contact might have a significant impact on resource access when the original bond between the nascent entrepreneur and his or her network partner is weak, the marginal benefits of higher relationship quality will diminish. At some point, network partners are already motivated enough to grant access to the resources they possess, which means that any further increase in relationship quality will not have an additional effect. Similarly, when working routines and shared norms of understanding are already well established and resource exchange is already very efficient, additional time spent on that specific relationship will not have an added impact on the availability of resources. In aggregate, we thus expect positive but diminishing resource returns with increasing relationship quality:

> **Hypothesis 2:** There will be a positive but concave relationship between the relationship quality in a nascent entrepreneur's network and resource access granted by network contacts.

Sample and Method

To identify a sample of nascent entrepreneurs for testing our hypotheses, we visited business start-up exhibitions in Germany. Here, we collected data from individuals indicating that they intended—as expected future owners—to found an economically and legally independent new venture as single entrepreneurs. Adapting Davidsson and Honig's (2003) operational definition for nascent entrepreneurs, we then limited our sample to those individuals who had already initiated at least one gestation activity. In particular, we limited our sample to those 379 individuals who indicated that they had either (1) already written a formal business plan or a less-formal description of their business model (79%), or (2) made, or were sure about making, a financial investment in the near future to set up their new venture (90%). In view of this selection procedure, we are confident that the respondents in our sample are nascent entrepreneurs. However, we certainly have to admit that the selection criteria we applied are not exactly the same as the ones used in the Panel Study of Entrepreneurial Dynamics (see, e.g., Gartner, Shaver, Carter, & Reynolds, 2004; Reynolds, 2011; Reynolds et al., 2004), which may be considered a limitation of our study.

To check for the representativeness of our sample, we compared it with the data from the German Socio-Economic Panel (GSOEP), a representative household panel survey often used for research on (nascent) entrepreneurs in Germany (see, e.g., Caliendo, Fossen, & Kritikos, 2009; Mueller, 2006; Schäfer & Talavera, 2009). We found a high degree of similarity between the nascent entrepreneurs in our sample and those within the GSOEP—for example, the 59.2% of male nascent entrepreneurs in our sample fairly matches the 58.3% observed within the GSOEP—and are thus confident that the results of our study are representative for nascent entrepreneurs in Germany.

Measures

An overview of the variables used in our analysis is reported in Table 2.

To capture our theoretical concepts, we relied on self-reports and single, tailor-made items. Even though this issue should be noted when considering our results, we are confident that this approach is appropriate. First, the main concepts that we addressed in our study are concrete attributes that may, thus, be validly measured using single items (Bergkvist & Rossiter, 2007, 2009). Second, previous research in entrepreneurship has yielded broad support for the reliability and validity of self-reported measures (Brush & Vanderwerf, 1992; Lechner, Dowling, & Welpe, 2006). Additionally, we checked for common method bias by performing a Harman's one-factor test. Because the results of our unrotated factor analysis show six factors with eigenvalues of more than one, with the maximum variance explained by a single factor being 15.9%, we are confident that common-method bias is not a significant issue in our study.

Dependent Variables. As described in the Theory and Hypotheses section, nascent entrepreneurs' networks are seen as an important source for access to financial capital, information/knowledge, and additional business-relevant contacts. We, thus, constructed three items and asked our respondents whether they had access to *financial capital*, *information/knowledge*, and *additional business contacts* via their network contacts. The respective resource variable took the value of 1 if the interviewee obtained that specific type of resource via his or her network, and 0 otherwise.

Table 2

Variables and Items

Variables	Items
Resource access	
Financial capital	Have you received financial support from your network contacts? [0 = no; 1 = yes]
Information/knowledge	Have you received relevant information/knowledge from your network contacts? [0 = no; 1 = yes]
Additional contact	Have you been introduced to additional business contacts by your network contacts? [0 = no; 1 = yes]
Network variables	
Network size	How many contacts do you currently have that you consider relevant for your founding project? [metric, in numbers]
Relationship quality	How many hours per week do you spend on average to maintain and further develop these contacts? [metric, in hours/total number of contacts]
Controls	
Gender	Your gender? [0 = female; 1 = male]
Marital status	Are you married? [0 = not married; 1 = married]
Age	How old are you? [metric, in years]
Education	Years of education? [metric, in years]
Industry experience	Do you have working experience in the industry in which you are trying to set up your new venture? [0 = no; 1 = yes]
Prior successful founding experience	Do you have prior experience in founding a new venture? If yes, have you previously succeeded in founding a new business? [0 = no; 1 = yes, successful]
Prior unsuccessful founding experience	Do you have prior experience in founding a new venture? If yes, have you previously failed in founding a new business? [0 = no; 1 = yes, successful]
Necessity entrepreneur	Are you unemployed or will you become unemployed in the near future if you do not switch into self-employment? [0 = no; 1 = yes]
Opportunity entrepreneur	Do you anticipate higher earnings as an entrepreneur? [0 = no; 1 = yes]
Support by family members and friends	How important is the support from your family/friends for the success of your founding project? [index of two 5-point scale items, anchored "1 = not important" to "5 = very important"]
Support by employer and colleagues	How important is the support from former employers and colleagues for the success of your founding project? [index of two 5-point scale items, anchored "1 = not important" to "5 = very important"]

Independent Variables. To capture the *size* of nascent entrepreneurs' *networks*, we followed the approach of Hansen (1995) and addressed all individuals who are somehow involved in setting up the nascent entrepreneur's business. Consequently, we adapted a measure constructed by Lechner et al. (2006), and asked our respondents for the total number of contacts they currently have and consider important for founding their business.

To measure *relationship quality*, we followed the approach used by Aldrich et al. (1987), as well as Aldrich and Reese (1993), and asked our respondents to indicate how much time they spent per week on developing and maintaining their network contacts. We then divided the total time spent by the number of network contacts to include in our analysis the average time spent per contact.

Controls. We also included several control variables in our analysis that might affect the demand for, or the availability of, resources via network contacts. We added *female* for two reasons: First, prior research indicates that there might be gender-specific barriers to entrepreneurial success and differences in the entrepreneurial process, as well as differences in the types of businesses started by male and female entrepreneurs (Langowitz & Minniti, 2007; Menzies, Diochon, Gasse, & Elgie, 2006; Murphy, Kickul, Barbosa, & Titus, 2007). Second, and even more important, some studies provide evidence for the notion that men and women may differ with respect to the composition of their networks, which may influence their respective opportunities to access resources via their network contacts. Specifically, Moore (1990) has shown that women tend to have a larger number and proportion of kin ties in their networks, a result that has been confirmed for female (nascent) entrepreneurs (Kwon & Arenius, 2010; Renzulli, Aldrich, & Moody, 2000). Additionally, we included *marital status* because being married indicates that respondents have a spouse who will probably be highly motivated to provide support and actively engage in the founding process (Brüderl & Preisendorfer, 1998; Ruef, Aldrich, & Carter, 2003; Sanders & Nee, 1996). Third, we controlled for *age* because of two underlying effects: as individuals get older, they tend to have more contacts (Renzulli et al.), and are also more likely to possess a higher stock of human and financial capital. Consequently, older nascent entrepreneurs may have less need for external resources than do younger ones (Parker, 2004).

We also decided to include different aspects of human capital into our analysis. First of all, we included *years of education* and *industry experience*, which may significantly impact nascent entrepreneurs' success in general (Samuelsson & Davidsson, 2009; Van Gelderen, Thurik, & Bosma, 2006), and the amount of knowledge and information they need and may have access to via their network contacts (Diochon, Menzies, & Gasse, 2008; Mosey & Wright, 2007; Yoo, 2000). As a third aspect of nascent entrepreneurs' human capital, we also included their prior founding experience, which has been found to exert a significant influence on entrepreneurial outcomes (Davidsson & Honig, 2003; Delmar & Shane, 2006; Rotefoss & Kolvereid, 2005; Van Gelderen et al.). Recognizing that founding experience in general does not necessarily indicate entrepreneurial expertise (Davidsson & Gordon, 2012), we included one measure for *prior successful experience with founding a new venture* and one measure for *prior unsuccessful experience*.

We also included two variables distinguishing the *opportunity entrepreneur*, who is "pulled" into the founding process by the perception of an entrepreneurial opportunity, and the *necessity entrepreneur*, who is usually "pushed" into nascent entrepreneurship by unemployment (Bosma, Acs, Autio, Coduras, & Levie, 2009; Schjoedt & Shaver, 2007). To control for a nascent entrepreneur's need to further develop their network, we also controlled for the relevance of support granted by two types of pre-existing network ties—*family members and friends*, and *former employers and colleagues*. As outlined earlier, the support granted by these pre-existing linkages may be especially important in the earlier stages of the founding process, and may therefore influence nascent entrepreneurs' need to develop and maintain additional network contacts (Elfring & Hulsink, 2003; Grandi & Grimaldi, 2003). Finally, we constructed a set of dummy variables representing the regions in which our interviews took place. In doing so, we accounted for potential differences in entrepreneurial environment, which may—as researchers in the field have argued (Busenitz et al., 2003; Mueller, 2006)—influence entrepreneurial activities.

Analytical Approach. Because our three resource access variables are binary, we considered multiple logistic regressions to be an appropriate model for our analysis. To test our hypotheses of diminishing resource returns associated with network size and relationship quality, we followed the method previously applied in entrepreneurship research by Colombo, Grilli, and Piva (2006), and described in detail by Wooldridge (2009), and included a linear as well as a squared term in our analysis. We then performed joint tests for significance to analyze whether the nonlinear model specifications were correct.

Results

The descriptive statistics for all variables used in our study, as well as their Pearson's correlations, are provided in Table 3.

The average nascent entrepreneur in our sample has a network with 14 contacts and spends a considerable amount of time, namely 0.85 hours per week, maintaining each of these contacts. Around 48% of our respondents were provided with financial resources by their network, 68% received informational support, and 64% have had access to additional business-relevant contacts via their network.

The correlation matrix reveals that nascent entrepreneurs' resource access through their network is significantly correlated with network size and—to a lesser degree—with relationship quality. The correlation also extends to several of our control variables, including age, education, and industry-specific experience. Additionally, the matrix shows that the support granted by family members and friends has a positive effect on access to financial capital and—to a lesser extent—on access to additional business-relevant contacts, whereas the support granted by former employers and colleagues has a significant impact on access to additional contacts and access to relevant knowledge and information. Since our independent variables are only moderately correlated, we are confident that multicollinearity is not an issue in our study.

The results of our logistic regressions, depicted in Table 4, predict access to financial capital and information/knowledge, as well as access to additional contacts as a function of network size, relationship quality, and our control variables. As a likelihood ratio chi-square test reveals, all of our models are significant.

With regard to our control variables, the regression results largely confirm the insights provided by the correlation matrix. They underline the importance of support granted by family members and friends, as well as former employers and colleagues. Additionally, some of our human capital variables and nascent entrepreneurs' ages have a significant influence on resource access. Finally, we see that nascent entrepreneurs who were unsuccessful in previously founding a new venture are less likely to receive financial support.

With respect to our hypotheses, our analyses confirm the proposed effects. Indicating a concave relationship, the coefficients of the linear terms for network size and relationship quality are positive, whereas the coefficients of the squared terms are negative. A joint test of significance further reveals that our nonlinear specifications of the regression models are correct. A Wald X^2 test rejects the null hypothesis that the coefficients of the quadratic terms are jointly equal to null for all three models tested (X^2 (2) = 12.70, 20.75, and 10.10, respectively).

In showing a positive but concave relationship between increasing network size and the probability of gaining access to financial capital, information/knowledge, and additional contacts via network members, our results provide confirming evidence for our first hypothesis. Similarly, and confirming our second hypothesis, the probability of getting access to different types of resources addressed in our study also increases at a decreasing rate with relationship quality.

To provide a clearer picture of our results, and to account for the fact that estimated probabilities in a nonlinear model strongly depend on the contribution of the covariates (Long & Freese, 2005; Mitchell & Chen, 2005), we then plotted the predicted probabilities for access to financial capital, relevant knowledge and information, and additional business contacts for three different types of entrepreneurs. Specifically, we estimated the predicted probabilities for (1) an "average" nascent entrepreneur in our sample by setting all control variables at their means (Type A), (2) an unmarried female nascent entrepreneur without prior founding and industry-specific experience who is "pushed" into

Table 3

Means, Standard Deviations (SD), and Pearson's Correlations

	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1. Financial capital	.48	.50	1																			
2. Information/knowledge	.68	.47	.34**	1																		
3. Additional contacts	.64	.48	.23**	.48**	1																	
4. Network size	14	21	.20**	.19**	.15**	1																
5. Relationship quality	.85	1.6	.02	.09	.07	−.18**	1															
6. Gender	.61	.49	.08	.03	−.03*	.04	−.13**	1														
7. Marital status	.33	.47	−.05	.03	.01	.01	−.05	−.01	1													
8. Age	37	9.7	−.10*	−.10*	.04	−.02	.06	−.14**	.38**	1												
9. Education	15	2.6	−.05	.10*	.07	.11*	−.08	−.08	−.01	.13**	1											
10. Industry experience	.57	.50	.02	.10*	.10*	.11*	.04	.03	.01	.05	−.02	1										
11. Founding exp. (success)	.11	.32	.06	.01	.03	.14**	−.05	−.01	.05	.07	.08	.14**	1									
12. Founding exp. (failure)	.07	.26	−.09	.02	.01	.02	−.05	−.01	.09	.06	−.11*	.17**	−.10*	1								
13. Necessity entrepreneur	.46	.50	−.05	−.11*	−.01	−.07	.01	−.08	.08	.31**	−.11*	.01	−.04	−.11*	1							
14. Opportunity entrepreneur	.58	.49	.04	−.02	.01	−.10*	.04	.10*	−.16**	−.22**	−.11*	.01	.01	−.01	−.04	1						
15. Support family/friends	3.1	1.2	.11**	.04	.08*	.05	.08	−.08*	.08*	−.07*	−.04	−.01	−.06	−.01	.04	.11**	1					
16. Support employer/colleagues	2.5	1.3	.07	.15**	.15**	.13**	−.02	.04	−.01	−.08*	.02	.05	−.06	.01	.02	.08*	.39**	1				
17. Environment I Stuttgart	.25	.44	−.01	−.06	−.02	−.06	−.05	−.01	.04	−.02	−.01	−.16**	−.04	−.01	−.12**	−.07*	.03	−.01	1			
18. Environment II Frankfurt	.08	.28	−.02	.01	−.01	−.01	−.01	−.06*	.07*	−.12**	−.12**	−.04	.02	.02	.02	.01	−.02	.01	−.18**	1		
19. Environment III Berlin	.23	.42	.05	.01	−.02	−.01	.10*	−.07*	−.09**	.02	.05	−.06	.08*	.01	.12**	.08*	−.11**	−.03	−.32**	−.17**	1	
20. Environment IV Bremen	.19	.39	−.08*	.02	.01	−.02	.08*	.01	−.04	−.01	−.04	.15**	−.04	.01	.04	−.03	.04	−.02	−.28**	−.15**	−.26**	1

** Correlation is significant at the 0.01 level (two-tailed); * correlation is significant at the 0.05 level (two-tailed).

Table 4

Robust Logit Estimation Results

	(1) Financial capital	(2) Information/ knowledge	(3) Additional contacts
Network size (number of contacts)	0.0481**	0.0642**	0.0496**
	(3.57)	(3.60)	(2.61)
Network size squared	–0.000197*	–0.000334**	–0.000281+
	(–2.34)	(–3.27)	(–1.71)
Relationship quality (average hours per contact/week)	0.661**	0.875**	0.515**
	(3.51)	(3.56)	(2.78)
Relationship quality squared	–0.0715**	–0.0590**	–0.0369**
	(–2.75)	(–3.36)	(–2.62)
Controls			
Gender (1 = male)	0.633*	0.274	0.0202
	(2.51)	(0.99)	(0.07)
Marital status (1 = married)	–0.141	0.507+	0.0960
	(–0.51)	(1.71)	(0.35)
Age (in years)	–0.00622	–0.0342*	–0.0121
	(–0.45)	(–2.29)	(–0.81)
Education (in years)	–0.0819+	0.0779	0.0618
	(–1.79)	(1.50)	(1.35)
Industry experience (1 = yes)	–0.0426	0.532*	0.400
	(–0.17)	(2.01)	(1.55)
Prior successful experience (1 = yes, success)†	0.169	–0.215	0.0807
	(0.44)	(–0.51)	(0.20)
Prior unsuccessful experience (1 = yes, failure)†	–0.851*	–0.0177	–0.324
	(–1.97)	(–0.04)	(–0.79)
Necessity entrepreneur (1 = yes)	0.174	–0.236	0.122
	(0.71)	(–0.89)	(0.47)
Opportunity entrepreneur (1 = yes)	0.0577	–0.126	–0.0930
	(0.24)	(–0.48)	(–0.37)
Support family and friends	0.222+	0.0570	0.271+
	(1.84)	(0.41)	(1.95)
Support employer/colleagues	0.168+	0.282*	0.253*
	(1.73)	(2.44)	(2.32)
Environment I: (1 = Stuttgart)‡	0.364	0.203	0.231
	(1.06)	(0.57)	(0.67)
Environment II: (1 = Frankfurt)‡	0.293	0.399	0.213
	(0.70)	(0.83)	(0.48)
Environment III: (1 = Berlin)‡	0.635+	0.449	0.169
	(1.72)	(1.09)	(0.44)
Environment IV: (1 = Bremen)‡	–0.510	0.218	–0.110
	(–1.43)	(0.55)	(–0.29)
Constant	–1.041	–1.717	–2.348*
	(–1.11)	(–1.53)	(–2.16)
Likelihood ratio chi-square test	47.67**	48.03**	46.91**
Pseudo R^2	0.130	0.158	0.115

t statistics in parentheses.
+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$
† No self-employment experience.
‡ Nurnberg

Figure 1

Network Characteristics and Resource Access (Type A)

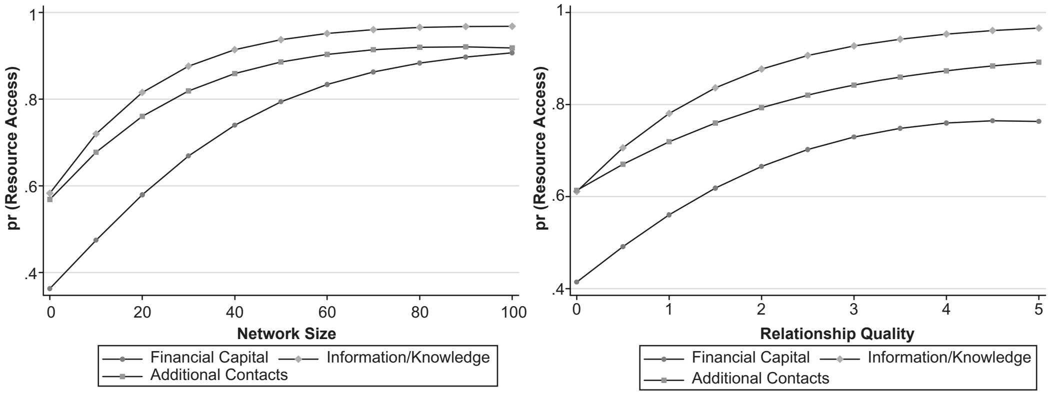
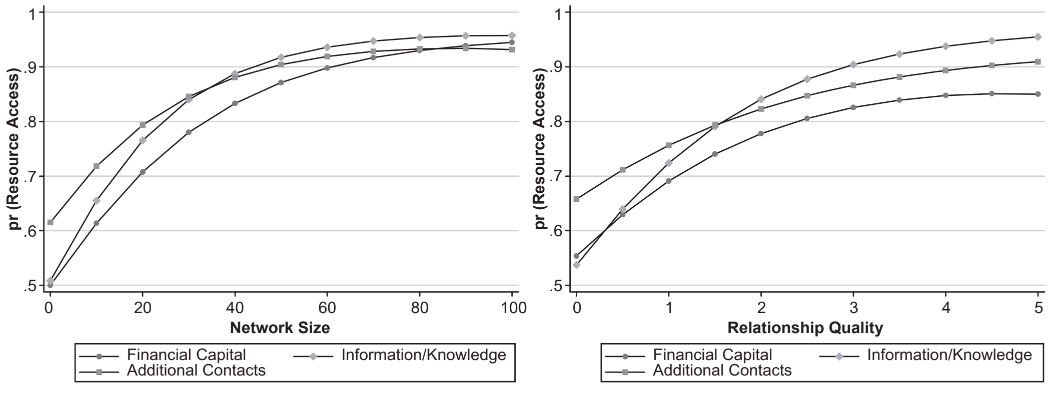


Figure 2

Network Characteristics and Resource Access (Type B)



self-employment (Type B), and (3) a married male with experience in successfully founding a new venture and industry experience who is “pulled” into nascent entrepreneurship (Type C). Figures 1–3 show the results.

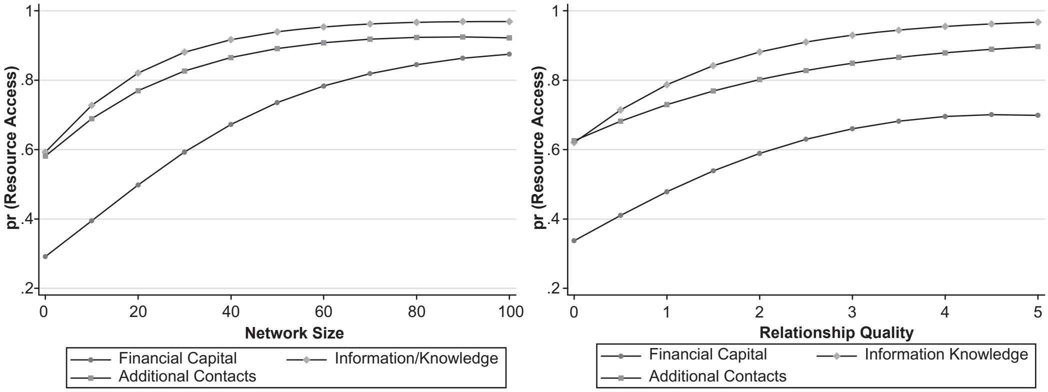
The figures reveal that whereas the relationships between our network variables and resource access differ only slightly by type of nascent entrepreneur, the slopes describing the probabilities of accessing the different types of resources vary significantly. While access to information and knowledge is quite easy to obtain, a much larger network and much more intense relationships are needed to access to additional business contacts and financial support.

Discussion

Based on the fact that research on the relationship between (nascent) entrepreneurs’ network characteristics and their success in founding and developing a new venture has so

Figure 3

Network Characteristics and Resource Access (Type C)



far produced conflicting results, our paper aimed at shedding more light on the bases of the widely proposed positive performance effect of network relationships. More specifically, we addressed the relationships between network characteristics and resource access, which have so far received very limited attention by empirical studies in the field (Street & Cameron, 2007; Stuart & Sorensen, 2007), and developed and tested the hypotheses that increasing network size and the quality of network relationships result in positive but diminishing resource returns for nascent entrepreneurs. Our analyses confirmed our hypotheses.

First, the results show that increasing network size enhances nascent entrepreneurs' opportunities to leverage resources such as financial capital, information and knowledge, and additional contacts through their network. However, our results also clearly show that extending the network leads to diminishing returns, meaning that the probability of gaining access to resources relevant for founding a new venture increases at a decreasing rate.

Analogously, our results also support our hypothesis that stronger relationship quality positively influences resource access. This finding underscores the idea that spending time together fosters trust and mutual understanding among network partners, as well as efficient resource exchange. Again, however, our results show that the marginal resource returns of increasing relationship quality are diminishing, which supports the argument that there is a point at which a further increase in relationship quality will not have a significant additional impact on exchange efficiency or on a partner's motivation to grant resource access.

In sum, these results may provide a partial explanation for why prior research addressing the relationship between network size and tie strength on the one hand, and entrepreneurial performance on the other, has led to ambiguous results. Because the resource returns from increasing the size of entrepreneurs' networks and the quality of their network relationships both diminish, one should not expect the overall performance effect to be linear. Instead, one has to consider two different but complementary effects: first, the diminishing resource returns that we deduced theoretically and confirmed by our empirical results, and second an additional cost effect. In particular, one also has to take into account that there are opportunity costs associated with developing and maintaining network relationships (Ebers & Grandori, 1997; Witt, 2004) when trying to predict the effects of establishing and maintaining network relationships on entrepreneurs' success. As also represented by the operationalizations used in our study, time and energy have to be invested to establish and maintain network relationships that may potentially provide access to resources helpful for founding and developing a new venture. This investment, however, comes with opportunity costs of less time available for other tasks relevant for new venture creation and development. When now considering this opportunity cost effect and the diminishing resource returns coming with increasing network size and relationship quality, a curvilinear relationship between these network variables and performance seems to be much more probable than the widely tested linear one.

From an empirical point of view, there is some evidence supporting this notion. First of all, Watson (2007) has shown that the frequency with which small- and medium-sized enterprise owners seek advice from different network sources first has a positive effect on their performance, but after a certain point may even hinder their survival and growth. Second, a closer look at the studies that have empirically addressed the relationship between network size and entrepreneurs' performance provides additional evidence for a curvilinear relationship between both variables. Specifically, it seems as if those studies reporting positive linear relationships between network size and entrepreneurs' performance are on average based on much smaller networks than those that find no such effect. Baum et al. (2000), Raz and Gloor (2007), as well as Hansen (1995), for example, refer to alliance or communication networks with 1.5, 2.8, and 8.3 network members on average. In contrast, the studies conducted by Batjargal (2003, 2005), which found no effect between network size and performance, are based on networks with an average size of 34.4 and 82, respectively. So even though we definitely acknowledge that these studies address different types of networks, and are based on very different methodological approaches, this observation may also provide some confirming evidence for the notion that further research should focus on testing for curvilinear relationships.

Besides showing diminishing returns for all three types of resources addressed, our results additionally indicate considerable variations in the relationships between network variables and access to the different types of resources. While access to information and knowledge is highly probable, even when a nascent entrepreneur has established only very few relationships at a low-intensity level, much more extensive networks and a higher level of relationship quality seem to be necessary to gain access to financial capital and additional contacts.

To explain this effect, it may be helpful to adopt the perspective of network partners. From their point of view, granting resource access is associated with certain costs that differ by resource type. While providing access to information and knowledge, for example, is almost cost-free, the situation is different when it comes to supporting a nascent entrepreneur with contacts or financial capital. In the case of financial capital, these expenses are quite apparent. As network exchange terms are expected to be much more favorable for nascent entrepreneurs than are market exchanges, the costs associated with providing access to financial capital are, for example, lower interest rates or more flexible payback arrangements that may negatively affect network alters' solvency. Providing contact support, however, also comes with significant costs since bringing people together always means giving up a bridging position between formerly unconnected actors (Burt, 1992). The network partner in the bridging position consequently loses the opportunity to exploit his or her middleman's position when bringing together the nascent entrepreneurs and his or her contact. Additionally, there are reputational risks involved when contacts are mediated because a contact broker is typically made responsible when a new contact turns out to be not as reliable or as trustworthy as expected (Coleman, 1988; Gulati & Gargiulo, 1999).

While this explanation for the different levels of network size and relationship quality needed to gain access to different types of resources is plausible, we are unfortunately not able to provide empirical evidence for this reasoning in our study. Since we gathered data only on the number of network relationships held and the average amount of time that entrepreneurs invest in their network relationships, we are unable to analyze the correspondence of the type of resource accessed and the time invested for a single network contact. Even though we acknowledge this limitation of our study as one that should be addressed by further research, we think that the results presented may still contribute to the discussion of the contingent value, and thus the performance effects, of network size and relationship quality in the field.

In showing that some entrepreneurs seeking access to a specific type of resource, such as additional business contacts, may still profit from extending their network or intensifying their network relationships, while others with a similar network that seek access to a different type of resource, such as knowledge and information, will predominantly suffer from the costs involved when doing so, our results contribute to the discussion on potential moderators of the effects of network size and relationship quality on entrepreneurs' success (see, e.g., Elfring & Hulsink, 2003; Hoang & Antoncic, 2003). Our results may, thus, help explain prior inconclusive findings in the field since they point to the necessity to account and control for the type of resource sought by entrepreneurs when addressing the performance effects of network characteristics.

Before concluding, we must note and discuss some additional limitations of our study. First of all, we clearly have to acknowledge that we are not able to precisely account for a nascent entrepreneur's position within his or her network. Considering the arguments presented in the Theory and Hypotheses section, such differences may influence the relationship between network size and relationship quality on the one hand, and resource access on the other. Following the view of Coleman (1988, 1990), one could, for example, argue that nascent entrepreneurs may profit from a denser network since a high level of interconnectedness facilitates cooperation and trust, and may thus enhance network partners' motivation to grant access to their resources. Consequently, one would expect a positive moderating impact of network density on the relationship between network size and relationship quality and resource access.

In contrast, one could propose that nascent entrepreneurs profit from sparser networks when following Burt's (1992, 2005) perspective. Specifically, one could argue that actors connecting otherwise unconnected network contacts, and thus holding a bridging position, are able to access more diverse resources and nonredundant information even when they only maintain relatively few network contacts. In view of these arguments, and the fact that the results of earlier studies on the direct effects of network density on entrepreneurs' performance have not been entirely conclusive, we explicitly want to encourage further research to address network structure as a potential moderator of the relationships between network size and relationship quality on the one hand, and resource access and performance on the other.

Closely connected to the aforementioned point, we also have to note that our focus on the entrepreneur's direct network keeps us from accounting for the indirect contacts held by an entrepreneur. We were, thus, not able to assess whether some nascent entrepreneurs rely on a small number of direct contacts that may function as gateways to larger networks of indirect ties (Kingsley & Malecki, 2004; Steier & Greenwood, 2000). For a number of reasons, however, we do not believe that capturing indirect network contacts would have significantly changed our results. According to Kim and Aldrich (2005), one has to expect such indirect network constellations to be rather short term. Especially when entrepreneurs have a sustaining interest in the resources provided by an indirect contact, we expect them to actively engage in forming a direct tie instead of steadily depending on a gateway (Reagans & Zuckerman, 2008; Steier & Greenwood). This argument is supported by the results of several research studies in the field. First, Chell and Baines (2000) and Julien, Andriambeloson, and Ramangalahy (2004) show that entrepreneurs usually do not consider institutions like universities or chambers of commerce—which are expected to serve as typical network hubs—as being important network contacts. Instead, they tend to rely on direct network relationships that they build up themselves. Second, existing research shows that the properties of an actor's network of direct and indirect ties are usually strongly positively correlated (Burt, 2007; Everett & Borgatti, 2005), and that indirect networks are less relevant for performance than originally expected (Burt).

Finally, we acknowledge that the cross-sectional design of our study may raise the issue of causality. Whereas a direct reversal of our interpretation does not seem very convincing, it is plausible to suppose that differences in network size and relationship quality may be driven by the motivation to compensate for shortfalls in specific types of resources—an argument that Brüderl and Preisendorfer (1998) recognize as the "network compensation hypothesis" (p. 224). This argument, however, would only challenge our research if the motivation to compensate for resource shortfalls influenced not only nascent entrepreneurs' network characteristics but also—independently of the first relationship—raised or lowered the probability that they have access to resources through their network. For several reasons, however, we do not think this is the case. First, the network compensation hypothesis has received only limited empirical support (Brüderl & Preisendorfer). Second, we controlled for several aspects of a nascent entrepreneur's human capital—which is the only variable for which the compensation hypothesis has been partially confirmed (Brüderl & Preisendorfer)—as well as for several other variables, such as regional environment and previous employment status, that not only indicate a nascent entrepreneurs' initial resource endowments but could also have an impact on their networks, and thus their opportunities to access resources through their network contacts.

Conclusion and Practical Implications

In this study, we investigated how two characteristics of nascent entrepreneurs' networks affect whether they have access to resources relevant for founding a new venture. Our results show positive but diminishing resource returns stemming from extending network size and increasing relationship quality. Additionally, we observe that the resource returns vary significantly by the type of resource accessed.

Considering these results, we believe that our paper extends knowledge of the performance effects of network size and relationship quality in the field of entrepreneurship in two ways: First, and in view of the empirical evidence provided for diminishing marginal resource returns that result when (nascent) entrepreneurs increase the size of their network or the quality or intensity of their network relationships, the conflicting findings provided by prior research may be partially explained by the fact that earlier research has predominantly tested for a linear performance effect of these network variables. When taking into account diminishing resource returns and opportunity costs, which are expected to come with developing and maintaining network relationships, however, it seems much more plausible to expect a curvilinear relationship between network characteristics and entrepreneurs' performance.

Second, the finding that the relationships between network size and relationship quality on the one hand, and access to start-up relevant resources on the other, vary considerably with the type of resource sought also contributes to our knowledge on the performance effects of these two network variables in the field. Specifically, this result indicates that networks of similar size and of similar relationship quality may have very different performance implications for entrepreneurs seeking access to different types of resources. It, thus, implies that further research addressing the performance effects of network variables in the field may have to account and control for the moderating influence of the type of resource entrepreneurs try to access via their network contacts.

We also strongly believe that the insights generated bear clear practical implications. On the one hand, our results underline the often-cited claim that networking activities may be valuable for (nascent) entrepreneurs (see, e.g., Davidsson & Honig, 2003; Miller, Besser, & Malshe, 2007), as investing time and energy in extending a network and increasing relationship quality may facilitate access to needed resources. On the other hand, our findings indicate that increasing the time and energy invested in networking leads to diminishing resource returns, and thus provides only limited support for the general notion that entrepreneurs should, in general, have large networks and spend much time with building and maintaining network relations (Greve, 1995). In fact, our results clearly run counter to calls for unlimited "power" networking—in terms of making as many contacts as possible—which is often promoted in popular entrepreneurship writing and sometimes even in entrepreneurship education. Instead, our results underscore the notion that entrepreneurs should focus on developing and maintaining a "network of essential ties" (Grandi & Grimaldi, 2003, p. 330), and carefully adjust the time they dedicate to these network relationships in terms of investing enough time to secure access to the needed resources but avoiding increasing the amount of time spent when there is no prospect of significant returns.

As for the optimal amount of time that nascent entrepreneurs should invest in extending their network and intensifying their network relationships, our results suggest that entrepreneurs consider the type of resource they want access to, and invest more time and energy on network development and maintenance when trying to obtain financial capital than when seeking access to additional information and knowledge. Answering the question of how much time entrepreneurs should actually spend on developing and maintaining network contacts in more detail, however, is not possible based on our study since it will largely depend on how seriously the respective resources are needed. Considering its practical relevance for (nascent) entrepreneurs, we think that this question should be further addressed by future research.

Our results also point to the fact that increasing network size and relationship quality are—at least to some extent—substitutable in their effects on resource access. This result not only means that nascent entrepreneurs have two options when trying to increase the probability of gaining access to a specific resource, but also implies that they should choose the one that enables them to most efficiently gain access to the needed resource. Consider, for example, a (nascent) entrepreneur who wants to get access to a specific piece of information. He or she may now pursue this goal by including additional contacts in his or her network, or by intensifying existing relationships to motivate existing network members to either share or gather and share the pieces of information required.

Our results also have clear implications for policy makers, who often subsidize networking events to facilitate (nascent) entrepreneurship and success (Casson & Giusta, 2007). On the one hand, they clearly confirm the general idea that networking events may be well suited to promote founding success since they provide an opportunity for entrepreneurs to develop their network, which may in turn increase their chances of accessing the resources needed to start a new venture. On the other hand, the results presented imply that merely providing networking opportunities might not be sufficient. Instead, policy makers should take into account that (nascent) entrepreneurs may need additional guidance to establish and maintain efficient networks, and thus consider providing them with opportunities to learn more about how to properly do so.

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